



Machine learning surrogates for Al_2O_3 – TiO_2 /water hybrid nanofluid battery thermal management under variable heat generation

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Abstract

Hybrid Al_2O_3 – TiO_2 /water nanofluids are promising coolants for lithium-ion battery thermal management systems, but evaluating their flowrate, heat-generation, and composition effects using CFD alone remains computationally demanding. This study develops machine-learning surrogate models for a liquid-cooled LG Chem 15 Ah pouch cell using a 72-case validated ANSYS Fluent dataset. The dataset combines a 28-case baseline matrix with a 44-case heat-generation extension covering 55–277.5 W. Three surrogate models multilayer perceptron (MLP), support vector regression (SVR), and convolutional neural network–long short-term memory (CNN-LSTM) were trained and evaluated using Reynolds number, heat generation rate, Al_2O_3 volume fraction, and TiO_2 volume fraction as inputs. The predicted outputs were the heat transfer coefficient and maximum hotspot temperature. The results show that the heat transfer coefficient is governed mainly by Reynolds number, while the maximum hotspot temperature is controlled by both Reynolds number and heat generation rate. Nanofluid composition provides a consistent but secondary correction to the thermal response, with the hybrid nanofluid giving the lowest hotspot temperature within the sampled range. SHAP attribution and bootstrap confidence intervals were used to interpret the MLP response and assess its sensitivity to dataset resampling, while constrained random sampling was used to map trends within the investigated design space. The resulting framework provides rapid, multi-output screening of hybrid-nanofluid BTMS operating conditions without requiring a new CFD calculation for each query, although predictions remain limited to the validated Reynolds-number, heat-load, and composition ranges.

Keywords Battery thermal management · Hybrid nanofluid · Machine learning surrogate · MLP · SVR · CNN-LSTM · SHAP · Monte Carlo · Bootstrap uncertainty quantification

Abbreviations

BTMS	Battery thermal management system
CFD	Computational fluid dynamics
CNN-LSTM	Convolutional neural network–long short-term memory
ML	Machine learning
MLP	Multilayer perceptron
RMSE	Root mean square error

SHAP	SHapley additive exPlanations
SVR	Support vector regression

List of symbols

D_h	Hydraulic diameter, m
h	Heat transfer coefficient, $\text{W m}^{-2} \text{K}^{-1}$
k	Thermal conductivity, $\text{W m}^{-1} \text{K}^{-1}$
Nu	Nusselt number
Pr	Prandtl number
Q	Heat generation rate, W
q''	Applied heat flux, W m^{-2}
Re	Reynolds number
$T_{\max}$	Maximum hotspot temperature, °C
T_{inlet}	Coolant inlet temperature, °C
T_{wall}	Area-averaged wall temperature, °C
μ	Dynamic viscosity, Pa s
ρ	Density, kg m^{-3}
$\Phi\text{Al}_2\text{O}_3$	Al_2O_3 volume fraction, vol%
ΦTiO_2	TiO_2 volume fraction, vol%

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Introduction

Lithium-ion batteries (LIBs) have become the cornerstone of modern electric vehicle (EV) powertrains and stationary energy storage owing to their high energy density, relatively low self-discharge rate, and long cycle life. Yet their electrochemical performance is tightly coupled to operating temperature. Sustained exposure beyond the safe window of 15–40 °C accelerates capacity fade, promotes electrolyte decomposition, and under extreme conditions initiates thermal runaway, a self-reinforcing exothermic cascade that poses direct safety risks to both vehicle occupants and infrastructure [1, 2]. Controlling cell temperature during high-rate discharge and fast-charging events therefore remains a central design constraint in EV battery engineering.

Active liquid cooling has emerged as the preferred thermal management strategy for high-energy-density packs, offering superior heat extraction rates compared to air-cooled and passive phase change material (PCM) designs [3, 4]. The thermal performance of any liquid-cooled BTMS is fundamentally constrained by the heat transfer coefficient of the circulating coolant. Conventional base fluids such as water and ethylene glycol exhibit limited thermal conductivity, which restricts heat extraction from cell surfaces under aggressive discharge conditions [5, 6]. This has prompted interest in nanofluids, which are stable suspensions of nanometer-scale particles in a base fluid and can serve as thermally enhanced coolant alternatives [7, 8].

The addition of metallic oxide nanoparticles such as Al_2O_3 , TiO_2 , and CuO to water at volume fractions below 5% yields measurable enhancements in thermal conductivity and convective heat transfer without proportionate increases in pumping power at optimized concentrations [9, 10]. Pak and Cho [11] established the foundational density and specific heat correlations for metallic oxide suspensions. More recently, hybrid nanofluids incorporating two or more nanoparticle species have been proposed to simultaneously tailor thermal conductivity, viscosity, and thermohydraulic performance beyond what single-component suspensions achieve [12, 13]. Kanti et al. [13] reported a 70.4% Nusselt number enhancement and a 46% reduction in total entropy generation for a 50:50 Al_2O_3 – TiO_2 water-based hybrid nanofluid under turbulent flow, while Venkateswarlu et al. [12] confirmed that hybrid nanofluids reduced prismatic battery surface temperatures by 4.12% relative to mono-nanofluid cooling [14]. These results establish hybrid nanofluids as credible high-performance coolants for liquid-cooled BTMSs, yet mapping their full design space through CFD alone is prohibitively expensive at the concentrations, flow rates, and geometric variants relevant to engineering practice [15].

Recent studies have increasingly combined hybrid nanofluids with modified mini-channel geometries. Sheikholeslami et al. [16] compared smooth, grooved, toothed, and pin–fin channels using an Fe_3O_4 –SWCNT/water hybrid nanofluid. Esmaeili and Sheikholeslami [17] examined bio-inspired fins under transient battery heat generation and evaluated heat transfer, entropy generation, and hydrothermal performance. More recently, Esmaeili et al. [18] coupled an Al_2O_3 – TiO_2 /water hybrid nanofluid with Fibonacci-inspired turbulators and compared smooth, parallel-flow, and alternating-counterflow arrangements. These studies demonstrate the combined influence of coolant composition and channel geometry on battery temperature control. Their conclusions, however, were obtained mainly from repeated, geometry-specific numerical calculations. The present study follows a complementary approach: the cooling geometry is held fixed, while Reynolds number, heat generation, and nanoparticle loading are varied to develop surrogate models that can screen the operating space without requiring a new CFD calculation for every design query.

Morali et al. [19] combined numerical simulations with an orthogonal design to examine the effects of ambient temperature, C-rate, nanoparticle concentration, and inlet velocity on the thermal response of prismatic batteries. Thawkar et al. [20] experimentally evaluated an Al_2O_3 –MWCNT/ethylene–glycol hybrid nanofluid in a helical-coiled pulsating heat pipe and demonstrated effective temperature regulation under discharge rates of 1C–3C. Atilgan and Yigit [21] used an experimentally validated three-dimensional CFD model to investigate the influence of cooling-plate channel configuration on pack temperature and temperature uniformity. These studies focused on direct parametric evaluation or cooling-hardware modification. They did not develop an interpretable surrogate for simultaneously predicting coolant-side heat transfer and battery hotspot temperature. The present study addresses this gap by using a validated CFD dataset to map the combined effects of Reynolds number, heat generation, and Al_2O_3 – TiO_2 composition through three machine-learning models, bootstrap resampling, and SHAP analysis.

Recent reviews have highlighted the increasing role of artificial intelligence in BTMS temperature prediction, design screening, and multi-objective optimization [22, 23]. However, they also identify continuing challenges associated with limited datasets, model interpretability, and validation against physically reliable thermal data. These concerns motivate the present use of a validated CFD dataset, cross-validated model comparison, bootstrap analysis, and SHAP-based interpretation. Machine-learning studies published in the host journal provide additional methodological context. Moolya et al. [24] compared several regression algorithms for predicting the thermal conductivity of Al_2O_3 /water nanofluids and used SHAP analysis to interpret the

predictions. Han et al. [25] combined numerical calculations with physics-informed machine learning for thermal-system optimization. These studies focused on nanofluid-property prediction or solar-thermal optimization rather than battery cooling. The present work extends this direction to a validated hybrid-nanofluid BTMS dataset and considers both heat-transfer coefficient and maximum hotspot temperature. Deep learning architectures including CNN-LSTM have recently been applied to nanofluid-cooled thermal systems [26] and ternary hybrid nanofluid BTMS configurations [27], extending the reach of data-driven surrogates to more complex thermal geometries. Multi-objective optimization of liquid cooling plate designs using deep learning surrogates has also demonstrated up to 95% reduction in computational cost relative to direct CFD search [4], confirming the practical engineering value of well-trained ML models as design tools. Yet surrogate-based cooling plate optimization studies have primarily used conventional coolants [28]; simultaneous treatment of nanofluid composition and flow conditions as continuous ML inputs remains uncommon [29]. Broader reviews have also documented the growing use of machine learning for predicting nanofluid thermophysical properties and heat-transfer behavior [30]. Garg et al. [31] combined Gaussian process regression with Monte Carlo sampling to evaluate uncertainty in nanofluid density and viscosity, while Naseri et al. [32] developed a hybrid data-driven model for thermal-conductivity prediction and examined its sensitivity using Monte Carlo analysis. Kam-suwan et al. [33] further demonstrated the use of artificial neural networks for predicting nanofluid properties and supporting heat-exchanger design calculations. These studies illustrate the value of data-driven property prediction and uncertainty analysis, but they do not provide an interpretable, multi-output surrogate for a hybrid-nanofluid battery thermal-management system.

Machine-learning-assisted design has also been extended to other thermal-energy systems. Sheikholeslami et al. [34] combined polynomial regression with differential evolution to optimize the geometry of a thin-film thermoelectric generator for maximum electrical power. Boora et al. [35] trained a neural-network surrogate using finite-volume results and coupled it with a Strength Pareto Evolutionary Algorithm for multi-objective optimization of a nanofluid solar receiver. At the system level, Sheikholeslami and Yarmohamad [36] investigated a hybrid-nanofluid-cooled photovoltaic–electrolyzer system, demonstrating the importance of coolant selection and thermal regulation in integrated energy conversion. Machine-learning surrogates have also been applied to heat exchangers for predicting thermal-performance indicators and heat-transfer coefficients [37, 38]. Together with the thermoelectric and solar-energy studies discussed above, these investigations demonstrate the wider value of surrogate models for computationally demanding

thermal systems. However, they do not address the combined effects of coolant flow, battery thermal load, and hybrid-nanofluid composition considered in the present study. The present work instead compares MLP, SVR, and CNN-LSTM models for two BTMS outputs and supplements the predictions with bootstrap uncertainty bounds and SHAP-based interpretation.

Existing studies generally follow two separate directions. Hybrid-nanofluid BTMS investigations rely mainly on repeated CFD calculations for particular channel geometries, whereas machine-learning studies frequently address isolated nanofluid properties or thermal systems outside battery cooling. Consequently, limited information is available on interpretable and uncertainty-aware surrogate models that simultaneously predict coolant-side heat transfer and battery hotspot temperature from a common, validated CFD dataset.

The contribution of the present study is methodological rather than the introduction of a new cooling geometry. A validated 72-case CFD dataset is used to compare three surrogate structures for two coupled BTMS outputs. The MLP was selected to represent nonlinear interactions between flow, heat generation, and composition. SVR was included because regularized kernel models are appropriate for small structured datasets, while CNN-LSTM was retained as a deep-learning benchmark to determine whether a more complex architecture provides any advantage for the tabular inputs. Bootstrap resampling evaluates the stability of prediction error, and SHAP analysis connects the surrogate response to the governing thermal mechanisms.

Once trained, the surrogates allow candidate operating conditions to be screened without repeating the CFD solution for every query. This offers a practical reduction in design-evaluation effort while preserving the physical trends established by the validated CFD model. The framework is intended for interpolation and screening within the sampled Reynolds-number, heat-load, and composition ranges; it is not proposed as a replacement for CFD or experimental confirmation outside that domain.

Table 1 compares the present framework with representative machine-learning studies in nanofluid heat transfer and battery thermal management.

Methodology

CFD dataset generation

The CFD dataset consisted of 72 ANSYS Fluent simulations for a liquid-cooled LG Chem 15 Ah pouch cell. The dataset was constructed in two stages. First, a 28-case baseline matrix was generated at a prescribed total thermal load of 277.5 W, to examine the effects of Reynolds number and nanoparticle loading. This matrix included four Reynolds

Table 1 Representative machine-learning surrogate studies related to nanofluid heat transfer and battery thermal management (UQ: uncertainty quantification; BTMS: battery thermal management system)

Study	Model(s)	Data source	Target output	UQ	Interpretability	Scope relative to present work
[42]	ANN, SVM, GPR	Experimental	Heat generation rate	None reported	Not reported	Composition effects not included
[13]	ANN	Experimental	Nusselt number; entropy generation	None reported	Not reported	No BTMS-specific composition mapping
[39]	MLP-ANN	Experimental	Viscosity	None reported	Not reported	Single-property prediction
[40]	MLP, LSSVM	Experimental	Nu, pressure drop	None reported	Not reported	Not applied to BTMS
[41]	GPR	Experimental	Thermophysical properties	None reported	Not reported	No BTMS application
[34]	Polynomial regression; differential evolution	COMSOL simulations validated against published experimental data	TEG electrical power	None reported	Not reported	No-BTMS geometry optimization
[35]	Neural network; Strength Pareto Evolutionary Algorithm	Validated finite-volume simulations	Nusselt number, friction factor, entropy generation, and efficiency	None reported	Not reported	Solar receiver; no BTMS application
Present work	MLP, SVR, CNN-LSTM	Validated CFD	(h) , $(T_{\max})$	Bootstrap resampling	SHAP attribution	Fixed 50:50 ratio; single-cell geometry

numbers, $Re = 2,300, 4,434, 7,000$, and $10,000$, and coolant cases consisting of water, Al_2O_3 /water at 1%, 2%, and 4%, and Al_2O_3 - TiO_2 /water hybrid nanofluid at 1%, 2%, and 4% total loading with a fixed 50:50 Al_2O_3 : TiO_2 ratio. Second, a 44-case heat-generation extension was added to evaluate whether the Reynolds-number and coolant-composition trends remained consistent across thermal loads from 55 to 277.5 W. This extension included water, Al_2O_3 /water, TiO_2 /water, and Al_2O_3 - TiO_2 /water hybrid nanofluid at 4% total loading. The combined 72-case dataset was used for final surrogate training and evaluation. Table 2 summarizes the operating conditions, coolant compositions, numerical

models, boundary conditions, and number of simulations used in the baseline matrix and heat-generation sweep.

Governing equations, boundary conditions and assumptions

Steady-state, incompressible Reynolds-averaged Navier–Stokes equations were solved together with conjugate heat transfer in the coolant and solid domains. The coolant was treated as a Newtonian, single-phase fluid with effective thermophysical properties. The conservation equations follow the ANSYS Fluent formulation and

Table 2 CFD parametric matrix and boundary conditions

Parameter	Baseline matrix	Heat-generation sweep
Cell dimensions	$160 \times 200 \times 5$ mm	$160 \times 200 \times 5$ mm
Heat generation	277.5 W	55, 111, 166.5, 222, 277.5 W
Reynolds number (Re)	2,300/4,434/7,000/10,000	3,200/4,434/5,800
Coolant compositions	Water; Al_2O_3 /water at 1%, 2%, 4%; Al_2O_3 - TiO_2 hybrid at 1%, 2%, 4% (50:50 ratio)	Water; Al_2O_3 /water, TiO_2 /water, Al_2O_3 - TiO_2 hybrid, each at 4% loading
Turbulence model	Realizable k- ϵ with enhanced wall treatment	Realizable k- ϵ with enhanced wall treatment
Nanofluid property model	Maxwell conductivity; Brinkman viscosity	Maxwell conductivity; Brinkman viscosity
Inlet temperature	20 °C	20 °C
Outlet condition	Pressure outlet (0 Pa gauge)	Pressure outlet (0 Pa gauge)
Simulation cases	28	44 (72 combined with baseline)

recent nanofluid-based battery thermal-management studies [43, 44].

The continuity equation is [43, 44].

$$\nabla \cdot (\rho_{\text{nf}} \mathbf{u}) = 0 \quad (1)$$

where ρ_{nf} and $\mathbf{u}$ are the effective coolant density and mean velocity vector, respectively.

The Reynolds-averaged momentum equation is [43, 44]

$$\rho_{\text{nf}}(\mathbf{u} \nabla) \mathbf{u} = -\nabla p + \nabla \cdot [(\mu_{\text{nf}} + \mu_t)(\nabla \mathbf{u} + \nabla \mathbf{u}^T)] \quad (2)$$

where p , μ_{nf} , and μ_t denote pressure, effective coolant viscosity, and turbulent viscosity, respectively.

The coolant energy equation is [43, 44]:

$$\rho_{\text{nf}} c_{p,\text{nf}}(\mathbf{u} \nabla T) = \nabla \cdot \left[\left(k_{\text{nf}} + \frac{\mu_t c_{p,\text{nf}}}{Pr_t} \right) \nabla T \right] \quad (3)$$

where $c_{p,\text{nf}}$, k_{nf} , and Pr_t are the effective specific heat, effective thermal conductivity, and turbulent Prandtl number, respectively. Viscous dissipation was neglected.

Steady heat conduction in the battery domain was described by the anisotropic solid-energy equation [43, 45, 46]

$$\nabla \cdot (\mathbf{K}_b \nabla T) + q''' = 0 \quad (4)$$

where $q''' = Q/V_b$ is the uniform volumetric heat-generation rate. The battery conductivity was represented by a diagonal tensor [45, 46]

$$\mathbf{K}_b = \text{diag}(k_{xx}, k_{yy}, k_{zz}) \quad (5)$$

The pouch cell was represented as a transversely isotropic solid. The 200 mm longitudinal direction was aligned with the y -axis, the 160 mm transverse direction with the x -axis, and the 5 mm through-thickness direction with the z -axis. Thermal conductivities of $k_{yy} = 15$, $k_{xx} = 15$, and $k_{zz} = 0.5 \text{ W m}^{-1} \text{ K}^{-1}$ were assigned. All off-diagonal components of the conductivity tensor were set to zero [43]. Thus, heat conduction was isotropic within the cell plane but anisotropic between the in-plane and through-thickness directions. The adopted values are within the ranges reported for layered pouch cells, for which the through-thickness conductivity is markedly lower than the in-plane conductivity [45, 46].

Temperature and normal heat flux were continuous at all coupled solid–fluid interfaces [43]:

$$T_f = T_s, -k_{\text{nf}} \nabla T_f \cdot \mathbf{n} = -\mathbf{K}_b \nabla T_s \cdot \mathbf{n} \quad (6)$$

The realizable $k - \epsilon$ model with enhanced wall treatment was used to represent turbulent transport [43, 47]. Turbulent viscosity was calculated from:

$$\mu_t = \rho_{\text{nf}} C_\mu \frac{k_t^2}{\epsilon} \quad (7)$$

where k_t is turbulent kinetic energy, ϵ is its dissipation rate, and C_μ is determined by the realizable formulation.

The Reynolds number was defined as [44]:

$$Re = \frac{\rho_{\text{nf}} U D_h}{\mu_{\text{nf}}} \quad (8)$$

where U is the mean inlet velocity and D_h is the channel hydraulic diameter.

A velocity inlet was imposed according to the specified Reynolds number, with a coolant inlet temperature of 20°C. A pressure outlet of 0 Pa gauge was applied at the channel exit. No-slip conditions were imposed on all coolant-contacting walls, while the external walls were treated as adiabatic. Temperature and heat flux were coupled at the solid–fluid interfaces. Gravity was disabled because forced convection was the dominant coolant-transport mechanism. The battery heat source was spatially uniform within the cell domain. The coolant properties were constant within each simulation and evaluated at 298 K.

Nanofluids were represented using a homogeneous single-phase approach. The nanoparticles and base fluid were assumed to be in local thermal equilibrium with no relative slip. Particle sedimentation, agglomeration and migration were neglected. These assumptions are appropriate for the low nanoparticle concentrations considered here but limit the model's ability to represent particle-scale transport effects [44].

Battery heat-generation model

The CFD calculations did not include an electrochemical submodel. Consequently, the Bernardi energy-balance formulation, which separates irreversible and reversible heat-generation contributions, was not evaluated directly [48]. Instead, the battery heat release was represented by a prescribed steady total thermal load, Q . Each load was converted to a volumetric source using $q''' = Q/V_b$, where the modeled battery volume was $V_b = 0.160 \times 0.200 \times 0.005 = 1.60 \times 10^{-4} \text{ m}^3$. The resulting source values are listed in Table 3.

In ANSYS Fluent, q''' was entered as a constant energy-source term in the battery solid cell zone. The source was spatially uniform and remained constant during each steady-state calculation, while no volumetric heat source was assigned to the coolant domain [43]. Reversible entropic heat and irreversible reaction and ohmic heat were not evaluated separately; Q was treated as a prescribed net thermal load. Charging and discharging transients, state-of-charge dependence and spatial variation in electrochemical heat generation were therefore outside the scope of the model. The five

Table 3 Total thermal loads and corresponding volumetric heat-generation rates applied to the battery

Total thermal load, (Q) (W)	Battery volume, V_b m ³	Volumetric heat source, q''' W m ⁻³
55.0	1.60×10^{-4}	3.4375×10^5
111.0	1.60×10^{-4}	6.9375×10^5
166.5	1.60×10^{-4}	1.040625×10^6
222.0	1.60×10^{-4}	1.3875×10^6
277.5	1.60×10^{-4}	1.734375×10^6

heat-load levels were introduced to examine the sensitivity of the cooling system and the machine-learning surrogates to thermal loading. Accordingly, they should be interpreted as prescribed thermal loads rather than direct electrochemical estimates of C-rate-dependent heat generation.

Nanofluid thermophysical properties

Nanofluids were modeled using the single-phase homogeneous mixture approach, valid for volume fractions below 5%. Effective thermal conductivity and viscosity were evaluated using the Maxwell [49] and Brinkman [50] correlations; density and specific heat from volume-weighted mixture relations. For the Al₂O₃–TiO₂ hybrid at 50:50 split, the Maxwell model was applied sequentially, yielding $k = 0.684 \text{ W m}^{-1} \text{ K}^{-1}$ at 4% total loading versus $k = 0.650 \text{ W m}^{-1} \text{ K}^{-1}$ for Al₂O₃/water alone a 5.3% conductivity advantage. Computed thermophysical properties for both fluid families are listed in Tables 4 and 5.

The nanofluids were represented using the homogeneous single-phase approach. For a total nanoparticle volume fraction ϕ , effective density and specific heat were evaluated using volume-weighted mixture relationships [44]:

$$\rho_{\text{nf}} = (1 - \phi)\rho_{\text{bf}} + \sum_i \phi_i \rho_i \quad (9)$$

$$c_{\text{p,nf}} = \frac{(1 - \phi)\rho_{\text{bf}}c_{\text{p,bf}} + \sum_i \phi_i \rho_i c_{\text{p,i}}}{\rho_{\text{nf}}} \quad (10)$$

where $\phi = \sum_i \phi_i$. For a single-component nanofluid, the summation contains one nanoparticle species; for the hybrid nanofluid, it includes Al₂O₃ and TiO₂.

Effective thermal conductivity was calculated using the Maxwell model [44, 49]:

$$k_{\text{nf}} = k_{\text{bf}} \frac{k_{\text{p}} + 2k_{\text{bf}} + 2\phi(k_{\text{p}} - k_{\text{bf}})}{k_{\text{p}} + 2k_{\text{bf}} - \phi(k_{\text{p}} - k_{\text{bf}})} \quad (11)$$

For the 50:50 Al₂O₃–TiO₂ hybrid, the Maxwell equation was applied sequentially using equal Al₂O₃ and TiO₂ volume fractions.

Effective viscosity was calculated using the Brinkman model [44, 50]:

$$\mu_{\text{nf}} = \frac{\mu_{\text{bf}}}{(1 - \phi)^{2.5}} \quad (12)$$

The nanofluid Prandtl number was calculated from its standard definition [43, 44]:

$$Pr_{\text{nf}} = \frac{\mu_{\text{nf}} c_{\text{p,nf}}}{k_{\text{nf}}} \quad (13)$$

Because the accuracy of the machine-learning surrogate models depends directly on the fidelity of the CFD-generated training dataset, the thermophysical-property models used to define the coolant inputs were evaluated against independent experimental measurements. Predictions from the Maxwell thermal-conductivity model were compared with the Al₂O₃/water and TiO₂/water measurements of Masuda et al. [51], while the Brinkman viscosity model was compared with the TiO₂/water measurements

Table 4 Thermophysical properties of Al₂O₃/water nanofluid at 298 K

Property	Water\0%	1%	2%	4%
Density /kg m ⁻³	998.0	1027.7	1057.4	1116.8
Specific heat /J kg ⁻¹ K ⁻¹	4182	4065	3953	3742
Thermal conductivity /W m ⁻¹ K ⁻¹	0.600	0.612	0.624	0.650
Dynamic viscosity /Pa·s	0.001000	0.001025	0.001051	0.001107
Prandtl number	6.970	6.808	6.658	6.373

Table 5 Thermophysical properties of Al₂O₃–TiO₂/water hybrid nanofluid at 298 K

Property	Water\0%	1%	2%	4%
Density /kg m ⁻³	998.0	1032.4	1066.8	1135.6
Specific heat /J kg ⁻¹ K ⁻¹	4182	4048	3920	3678
Thermal conductivity /W m ⁻¹ K ⁻¹	0.600	0.620	0.641	0.684
Dynamic viscosity /Pa·s	0.001000	0.001025	0.001051	0.001107
Prandtl number	6.970	6.692	6.427	5.953

of Duangthongsuk and Wongwises [52]. Comparisons were performed using normalized property ratios at corresponding particle volume fractions (Figs. 1 and 2; Tables 6 and 7).

The Maxwell model reproduced the TiO₂/water thermal-conductivity measurements closely, producing a MAPE of 0.42% and a maximum deviation below 1%. Larger deviations were observed for Al₂O₃/water, for which the model

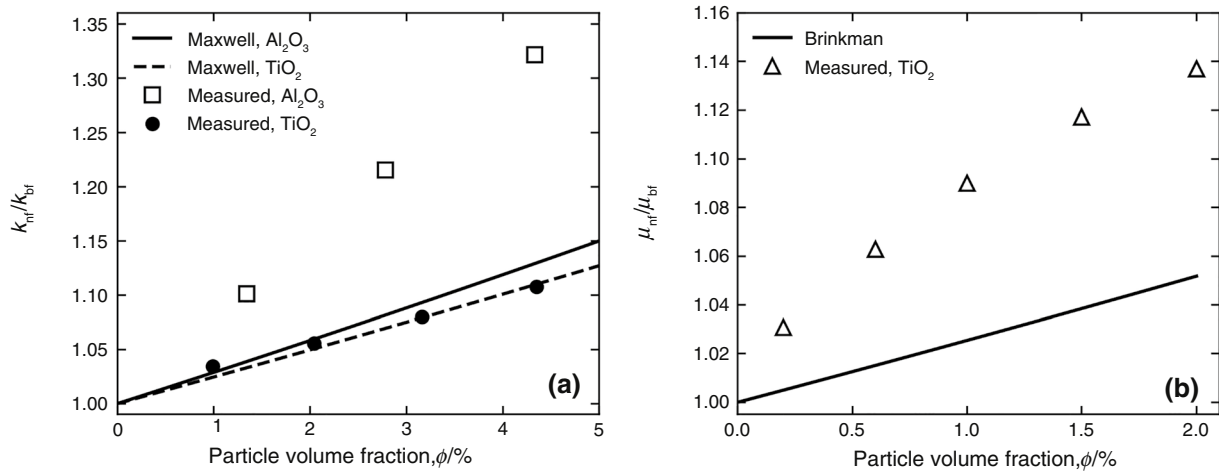


Fig. 1 Comparison of classical thermophysical-property models with independent experimental measurements: (a) normalized thermal conductivity of Al₂O₃/water and TiO₂/water nanofluids predicted

using the Maxwell model and (b) normalized viscosity of TiO₂/water nanofluid predicted using the Brinkman model

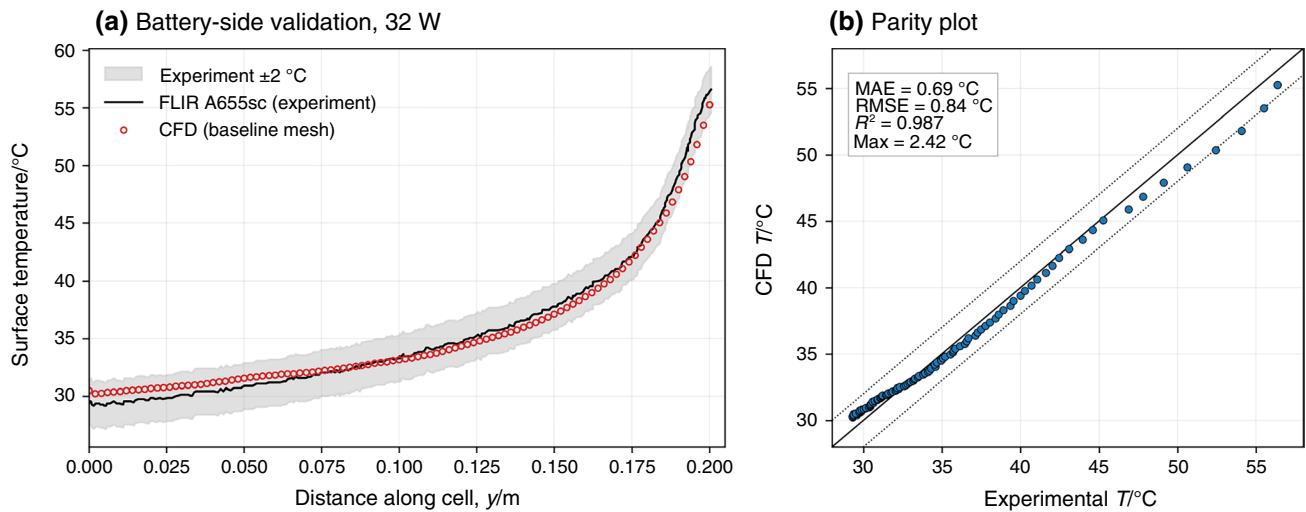


Fig. 2 Battery-side temperature validation at a total heat input of 32 W: (a) experimental and CFD surface-temperature profiles, including the FLIR accuracy range ($\pm 2^\circ\text{C}$); (b) parity plot of CFD and experimental temperatures

Table 6 Quantitative comparison of the thermophysical-property models with experimental measurements

Property ratio	Nanofluid	Model	Experimental source	N	MAE	RMSE	MAPE\%	Maximum deviation\%
k_{nf}/k_{bf}	Al ₂ O ₃ /water	Maxwell	Masuda et al. [51]	3	0.130	0.140	10.41	14.56
k_{nf}/k_{bf}	TiO ₂ /water	Maxwell	Masuda et al. [51]	4	0.004	0.006	0.42	0.96
μ_{nf}/μ_{bf}	TiO ₂ /water	Brinkman	Duangthongsuk and Wongwises [52]	5	0.060	0.064	5.48	7.49

Table 7 Grid-independence analysis for water at $Re = 10,000$, $Q = 277.5$ W and $T_{inlet} = 20^\circ\text{C}$

Grid	Elements	$T_{max}\backslash^\circ\text{C}$	$\Delta T_{max}\backslash\%$	$h\backslash\text{W m}^{-2}\text{K}^{-1}$	$\Delta h\backslash\%$
Grid 1 (coarse)	159,579	42.300	—	6641.74	—
Grid 2 (selected)	285,698	41.845	1.076	6570.32	1.075
Grid 3 (fine)	405,367	41.839	0.014	6571.24	0.014

underpredicted the measured conductivity enhancement with a MAPE of 10.41%. The Brinkman model also underpredicted the measured TiO_2 /water viscosity, although its MAPE remained 5.48%. These discrepancies are attributed to particle-size, aggregation, interfacial-layer and temperature effects that are not represented explicitly in the classical Maxwell and Brinkman formulations. Because directly comparable experimental data were unavailable for the 50:50 Al_2O_3 – TiO_2 hybrid formulation, the sequential Maxwell treatment of the hybrid coolant remains a modeling assumption and is identified as a limitation.

Mesh independence and solver settings

A three-level grid-independence study was performed for the forced-convection water-cooling model at $Re = 10,000$, a heat-generation rate of 277.5 W, and an inlet temperature of 20°C . The coarse, medium and fine grids contained 159,579, 285,698 and 405,367 elements, respectively. The maximum battery temperature, T_{max} , which also represents the hotspot temperature, and the area-averaged heat-transfer coefficient, h , were monitored as the grid-convergence quantities. Refinement from the coarse to the medium grid changed T_{max} and h by 1.076% and 1.075%, respectively. Further refinement from the medium to the fine grid changed both quantities by only 0.014%. Although the fine grid contained 41.9% more elements than the medium grid, it produced no meaningful improvement in either output. Therefore, the medium grid containing 285,698 elements was selected for all 72 production simulations. The realizable $k - \epsilon$ turbulence model with enhanced wall treatment, SIMPLE pressure–velocity coupling, and second-order upwind discretization were applied consistently to all grids. Convergence was declared when the energy residual decreased below 10^{-6} and all other residuals decreased below 10^{-4} .

Experimental validation

The battery-side thermal model was quantitatively evaluated against steady-state FLIR A655sc infrared-thermography measurements obtained for a $160 \times 200 \times 5$ mm pouch cell under a total heat input of 32 W [53]. The experiment was conducted at an ambient temperature of 300 K, and the cell

Table 8 Forced-convection consistency check between CFD-derived and Gnielinski-predicted Nusselt numbers for water-cooled cases

Re	$Q\backslash\text{W}$	Nu_{CFD}	Nu_{Gn}	Error\%
3200	55	24.85	24.34	2.10
4434	111	32.30	35.48	8.96
5800	166.5	40.10	46.98	14.64
7000	277.5	50.75	56.60	10.33
10,000	277.5	65.70	79.36	17.21

surface was coated matte black to improve emissivity control. CFD and experimental temperatures were paired at 354 locations along the 200 mm vertical cell axis. The comparison produced an MAE of 0.688°C , an RMSE of 0.838°C , a mean bias of -0.195°C , and $R^2 = 0.9869$. Overall, 96.9% of the CFD predictions were within the nominal FLIR measurement accuracy of $\pm 2^\circ\text{C}$. The maximum absolute deviation was 2.417°C at $y = 0.1956\text{m}$, near the tab region where localized heating produces the steepest temperature gradient.

Collectively, the thermophysical-property assessment, grid-independence study and battery-side experimental comparison quantify three sources of fidelity in the CFD-generated training dataset: coolant-property specification, numerical discretization and battery-temperature prediction. Together, these checks provide evidence for the numerical and physical fidelity of the CFD dataset used to train the surrogate models. The resulting surrogate predictions are therefore restricted to the Reynolds-number, heat-generation and nanoparticle-concentration ranges represented by the 72 CFD cases.

Forced-convection consistency check

Since the available experimental validation was conducted under 32 W natural convection, an additional forced-convection consistency check was performed for the water-cooled cases. The CFD-derived heat transfer coefficient was converted to a Nusselt number using $Nu_{\text{CFD}} = hD_h/k$, where D_h is the hydraulic diameter and k is the thermal conductivity of water. The resulting values were compared with the Gnielinski turbulent internal-flow correlation over the simulated Reynolds-number range [54]. As shown in Table 8, the CFD-derived Nusselt numbers agreed with the correlation within 2.10–17.21%, with an average deviation of 10.65%. This agreement indicates that the forced-flow heat-transfer coefficients predicted by the CFD model are physically consistent with established internal-flow behavior.

The remaining deviations may be attributed to developing flow, the non-circular coolant passage, nonuniform wall temperature, and conjugate heat transfer in the CFD model, whereas the Gnielinski correlation assumes fully developed flow in a smooth duct. The baseline dataset also included a

condition at $Re = 2300$, which lies near the laminar–turbulent transition. This condition was excluded from the Gnielinski comparison because it falls outside the applicable turbulent range of the correlation. The realizable k – ϵ model was retained at this condition to maintain a consistent numerical procedure throughout the dataset. Therefore, the result at $Re = 2300$ should be interpreted as a transitional-flow approximation and may carry greater model-form uncertainty than the fully turbulent cases [43, 47, 54].

Input features and output variables

Four input features were used for surrogate modeling: Reynolds number Re , heat generation rate Q , Al₂O₃ nanoparticle volume fraction $\phi_{\text{Al}_2\text{O}_3}$, and TiO₂ nanoparticle volume fraction ϕ_{TiO_2} . The target outputs were the area-averaged heat transfer coefficient h , maximum hotspot temperature T_{max} .

The area-averaged heat-transfer coefficient was calculated from the applied heat flux and wall-to-inlet temperature difference [43, 54]:

$$h = \frac{q''}{T_{\text{wall}} - T_{\text{inlet}}} \quad (14)$$

where q'' is the applied wall heat flux, T_{wall} is the area-averaged cell surface temperature, and T_{inlet} is the coolant inlet temperature.

Machine learning models

Three model architectures were trained and compared using the same CFD dataset and cross-validation framework. Figure 3 summarizes the CFD–ML workflow used in this study. The workflow begins with CFD dataset generation and validation, followed by preprocessing, surrogate-model training, model evaluation, bootstrap uncertainty quantification, SHAP attribution, constrained design-space mapping, and engineering interpretation.

Multilayer perceptron (MLP)

A feed-forward MLP with two fully connected hidden layers was constructed: 64 neurons (ReLU) and 32 neurons (ReLU), with a single linear output neuron. Training used the Adam optimizer at a learning rate of 0.001 with MSE loss. Early stopping with patience of 50 epochs on held-out fold validation loss was applied to limit overfitting.

Support vector regression (SVR)

An SVR model with RBF kernel was implemented. Regularization parameter C and kernel bandwidth γ were selected by grid search over $C \in \{1, 10, 100, 1000\}$ and $\gamma \in \{0.001, 0.01, 0.1, 1\}$, using mean cross-validation RMSE as the criterion. Optimal values of $C = 100$ and $\gamma = 0.1$ were applied uniformly across all five folds.

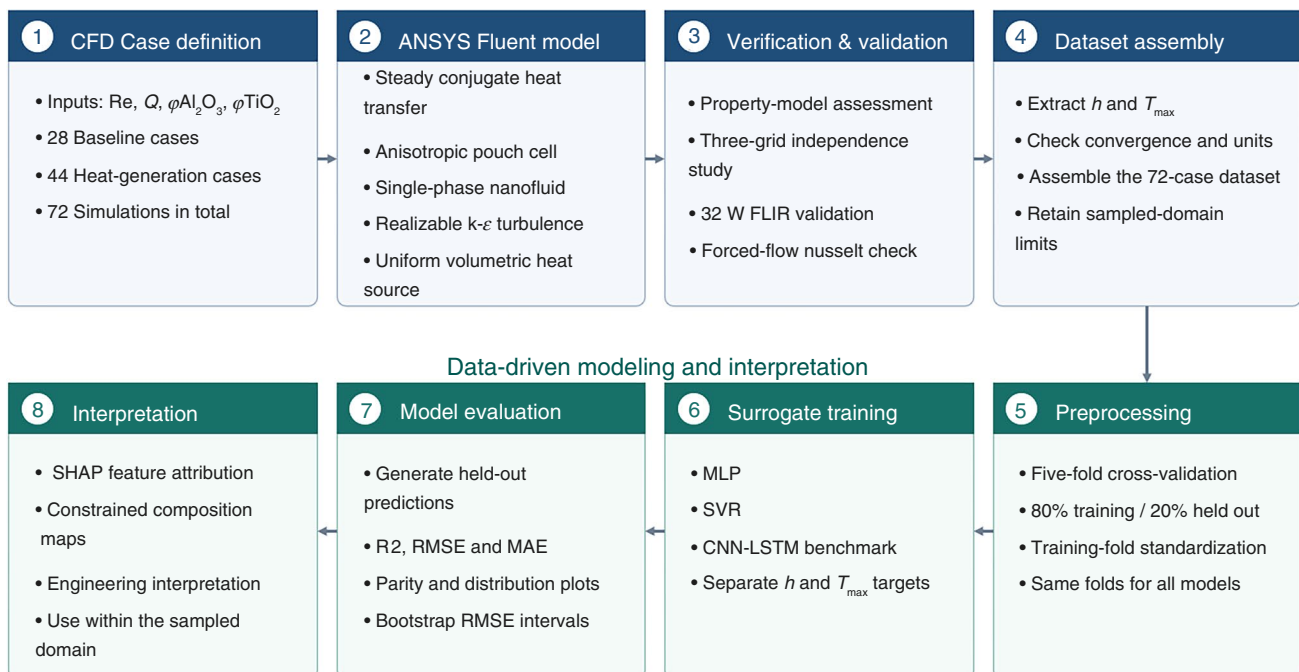


Fig. 3 Complete CFD-to-machine-learning workflow used for CFD verification and validation, dataset generation, surrogate-model training, prediction, uncertainty assessment, and engineering interpretation

CNN-LSTM

The CNN-LSTM was included as a comparative deep-learning benchmark. The four standardized input features were reshaped as a short one-dimensional sequence and processed using convolutional and LSTM layers, followed by a linear regression output. Separate models were trained for h and $T_{\max}$. The model was evaluated using the same five-fold cross-validation partitions used for the MLP and SVR. Because the dataset contained only 72 cases and the inputs were tabular rather than temporal, the CNN-LSTM was treated as a benchmark rather than the primary surrogate.

Model evaluation

All three models were assessed using fivefold cross-validation. In each fold, approximately 80% of the 72 cases were used for training and the remaining 20% were used for testing, corresponding to 57–58 training cases and 14–15 test cases per fold. Model performance was evaluated using the coefficient of determination, root-mean-square error, and mean absolute error, following the definitions given by Pedregosa et al. [55]:

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y}_i)^2} \quad (15)$$

$$RMSE = \sqrt{\left[(1/n) \sum (y_i - \hat{y}_i)^2 \right]} \quad (16)$$

$$MAE = (1/n) \sum |y_i - \hat{y}_i| \quad (17)$$

Table 9 summarizes the performance metrics calculated from the combined held-out predictions generated during fivefold cross-validation. The MLP achieved the best performance for both outputs. For h , the MLP produced $R^2 = 0.983$, an RMSE of $174.535 \text{ W m}^{-2} \text{ K}^{-1}$, and an MAE of $122.472 \text{ W m}^{-2} \text{ K}^{-1}$. For $T_{\max}$, it achieved $R^2 = 0.946$, an RMSE of $3.800 \text{ }^\circ\text{C}$, and an MAE of $2.694 \text{ }^\circ\text{C}$. CNN-LSTM

Table 9 Performance of the surrogate models based on the combined fivefold cross-validated predictions

Output	Model	(R^2)	RMSE	MAE
h	MLP	0.983	$174.535 \text{ W m}^{-2} \text{ K}^{-1}$	$122.472 \text{ W m}^{-2} \text{ K}^{-1}$
h	CNN-LSTM	0.977	$214.280 \text{ W m}^{-2} \text{ K}^{-1}$	$160.673 \text{ W m}^{-2} \text{ K}^{-1}$
h	SVR	0.953	$303.627 \text{ W m}^{-2} \text{ K}^{-1}$	$237.186 \text{ W m}^{-2} \text{ K}^{-1}$
$T_{\max}$	MLP	0.946	$3.800 \text{ }^\circ\text{C}$	$2.694 \text{ }^\circ\text{C}$
$T_{\max}$	CNN-LSTM	0.833	$6.297 \text{ }^\circ\text{C}$	$4.665 \text{ }^\circ\text{C}$
$T_{\max}$	SVR	0.494	$12.775 \text{ }^\circ\text{C}$	$10.208 \text{ }^\circ\text{C}$

ranked second, while SVR produced the largest errors, particularly for $T_{\max}$. Therefore, the selection of MLP as the primary surrogate was based on the numerical cross-validation results rather than visual inspection of the parity plots alone.

SHAP feature attribution analysis

SHAP values were calculated for the trained MLP using exact interventional Shapley attribution. Because the model contained four input features, all 16 possible feature subsets were evaluated using the complete 72-case dataset as the empirical background distribution. Mean absolute SHAP values were used to rank the global influence of Reynolds number, heat-generation rate, Al_2O_3 volume fraction, and TiO_2 volume fraction. Signed SHAP values were used to determine whether each feature increased or decreased the standardized model output. Because these attributions explain the behavior of the trained surrogate, they were interpreted as model-dependent associations rather than causal relationships [56, 57].

Monte Carlo design space exploration

Monte Carlo mapping was used as an exploratory visualization of the surrogate response across the Al_2O_3 – TiO_2 composition plane at two Reynolds numbers under the same heat-generation rate. The design-space comparisons were performed at $\text{Re}=4,500$ and $\text{Re}=9,000$ for prescribed thermal loads of 80 and 180 W. These values lie within the overall sampled ranges but do not coincide with all original CFD design levels; therefore, the resulting maps represent surrogate interpolation in Reynolds number and thermal load. The CNN-LSTM surrogate was used only for qualitative contour visualization because it provides continuous predictions over the sampled input plane. For each operating condition, 10,000 candidate compositions were sampled within the physically constrained region $\phi_{\text{Al}_2\text{O}_3} + \phi_{\text{TiO}_2} \leq 4\%$. This constraint prevents extrapolation to total nanoparticle loadings above the maximum concentration used in the CFD matrix. The predicted values were then interpolated onto a regular grid and plotted as filled contours. Since the CFD training matrix contains only the fixed 50:50 Al_2O_3 : TiO_2 hybrid ratio, predictions away from this diagonal should be interpreted as extrapolative surrogate trends rather than independently validated coolant compositions. In the present study, Monte Carlo sampling was used to explore the constrained design space rather than to propagate probabilistic input uncertainty. Accordingly, the contour maps represent surrogate response trends and should not be interpreted as probability distributions or confidence regions. Statistical prediction uncertainty was assessed separately through bootstrap resampling.

Bootstrap uncertainty quantification

Prediction uncertainty for the three data-driven surrogates was characterized using bootstrap resampling of the CFD dataset [58]. For each bootstrap replicate, the model was retrained using a resampled training set and evaluated using out-of-bootstrap observations. The 2.5th and 97.5th percentiles of the resulting RMSE distribution were used to define the 95% bootstrap interval. Separate bootstrap analyses were conducted for the heat-transfer-coefficient and maximum-hotspot-temperature predictions.

Computational environment and architectural reproducibility

All data-driven surrogate models were compiled, trained, and evaluated using Python and MATLAB programming environments. The numerical computations were executed on a desktop workstation powered by an AMD Ryzen 7 5700X3D processor and an NVIDIA GeForce RTX 3060 Ti graphics processing unit. Due to the compact nature of the physics-informed training matrix, the computational cost of model training was remarkably low: MLP and SVR cross-validation folds required less than 5 s to train. Once trained, evaluating any prospective coolant composition configuration across the full surrogate design envelope required less than 10 ms of inference time, achieving a massive computational speedup compared to the standard multi-hour ANSYS Fluent CFD simulations.

Results and discussion

The results are discussed using the combined 72-case CFD dataset. The 28-case baseline matrix captures Reynolds-number and nanoparticle-loading effects at 277.5 W, while the 44-case extension adds heat-generation variation from 55 to 277.5 W. This combined dataset allows the surrogate models to evaluate the relative influence of flow rate, thermal load, and coolant composition on h , $T_{\max}$, and ϕ .

Effect of Reynolds number and heat generation rate on heat transfer coefficient

Figure 4 compares the CFD-derived heat transfer coefficient with the corresponding ML predictions for CNN-LSTM, MLP, and SVR. The results show that the models capture the main trend in h , although the prediction scatter differs between architectures. The MLP predictions remain closest to the parity line, while CNN-LSTM and SVR show larger deviations in some regions of the dataset. This indicates that h can be represented by the surrogate framework, but model selection remains important for prediction accuracy.

Figure 5 shows that the MLP most closely preserves the central tendency and spread of the CFD-derived h values. Figure 6 provides the corresponding bootstrap RMSE distribution. The mean RMSE was approximately $186 \text{ W m}^{-2} \text{ K}^{-1}$, while the 95% bootstrap interval extended from approximately 132 to $242 \text{ W m}^{-2} \text{ K}^{-1}$.

This interval describes the sensitivity of the surrogate to resampling of the available CFD cases. Its width indicates that the prediction error increases when influential operating conditions near the limits of the dataset are omitted from a

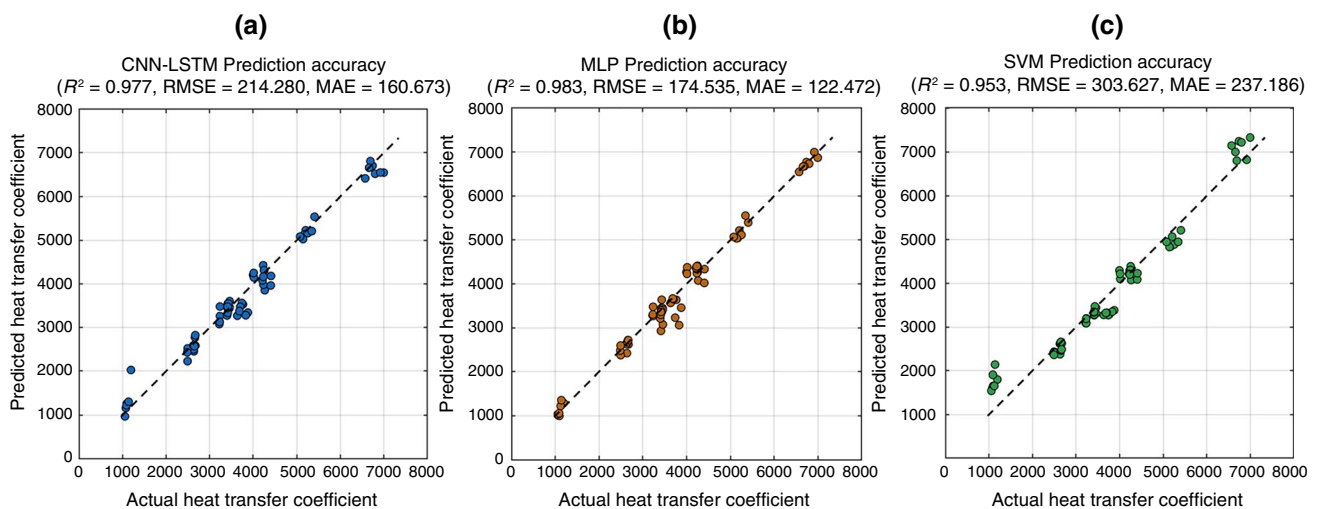


Fig. 4 Cross-validated CFD-derived versus ML-predicted heat transfer coefficient h : (a) CNN-LSTM, (b) MLP, and (c) SVR. The dashed line indicates perfect agreement

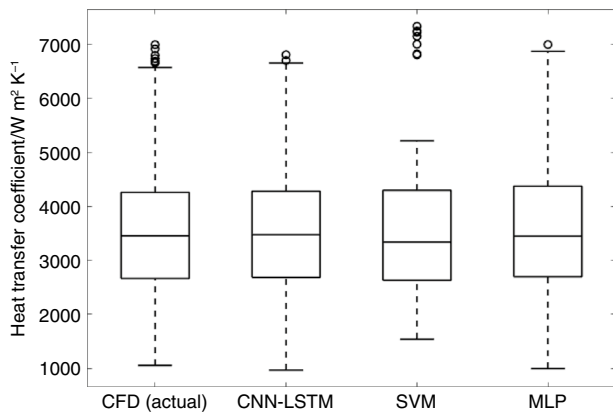


Fig. 5 Boxplot comparison of CFD-derived and ML-predicted heat transfer coefficient h values for CNN-LSTM, SVR, and MLP

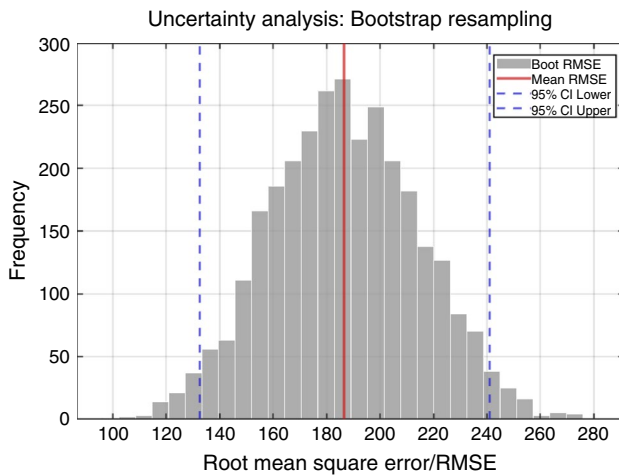


Fig. 6 Bootstrap RMSE distribution for MLP heat-transfer-coefficient predictions. Dashed lines indicate the 95% confidence interval

bootstrap sample. Therefore, the surrogate is suitable for rapid screening within the sampled domain, but designs close to the Reynolds-number or heat-load limits should be confirmed by CFD.

Monte Carlo mapping of heat transfer coefficient

Figures 7 and 8 show that Reynolds number produces a substantially larger change in h than either heat generation or nanoparticle composition. Higher Re increases the coolant momentum and suppresses the growth of the thermal boundary layer along the cooled surface. The resulting increase in the wall-temperature gradient raises the Nusselt number and the area-averaged heat-transfer coefficient. Heat generation has little effect on h at a fixed flow conditions because the applied heat flux and wall-to-coolant temperature difference increase approximately proportionally.

Increasing nanoparticle loading produces a smaller enhancement through the effective thermal conductivity of the coolant. At 4% total loading, the calculated conductivity of the hybrid nanofluid increased from 0.600 to 0.684 $\text{W m}^{-1} \text{K}^{-1}$, corresponding to a 14% increase relative to water. However, the viscosity also increased from 0.001000 to 0.001107 Pa s. Therefore, the thermal benefit cannot be treated as a thermohydraulic optimum because pumping power and pressure drop were not evaluated in the present study.

Effect of Reynolds number and heat generation rate on maximum hotspot temperature

Figure 9 compares the CFD-derived maximum hotspot temperature with the corresponding ML predictions for CNN-LSTM, MLP, and SVR. The temperature prediction problem is more sensitive than the h prediction problem because $T_{\max}$ depends directly on both heat generation rate and convective heat removal. The MLP predictions follow the perfect-agreement line more closely, while CNN-LSTM and SVR show larger scatter, especially for the higher-temperature cases. This indicates that the surrogate framework can predict the main thermal response, but high-temperature cases require careful interpretation.

Figure 10 compares the distribution of CFD-derived and ML-predicted $T_{\max}$ values. This comparison is important because a model may show acceptable average error while still underpredicting or overpredicting the high-temperature cases. Preserving the upper-temperature range is especially important for battery thermal-management design, where local hotspot underprediction can lead to unsafe design conclusions.

Figure 11 shows that the MLP hotspot-temperature error is more sensitive to the available training cases than the heat-transfer-coefficient error. The mean bootstrap RMSE was approximately 4.2 °C, with a 95% interval of approximately 2.9–5.8 °C. The upper part of this interval is significant for BTMS design because an underprediction of several degrees may change whether a proposed operating condition satisfies a temperature limit. Consequently, surrogate predictions close to a specified safety threshold should be checked using the validated CFD model. The bootstrap interval represents dataset-resampling uncertainty only; it does not include uncertainty in coolant properties, CFD model form, boundary conditions, or experimental measurements.

SHAP-based interpretation of the MLP predictions

Figure 12 presents the SHAP distributions for the MLP predictions of the heat-transfer coefficient and maximum hotspot temperature. Reynolds number is the dominant feature for the heat-transfer coefficient. Higher Reynolds

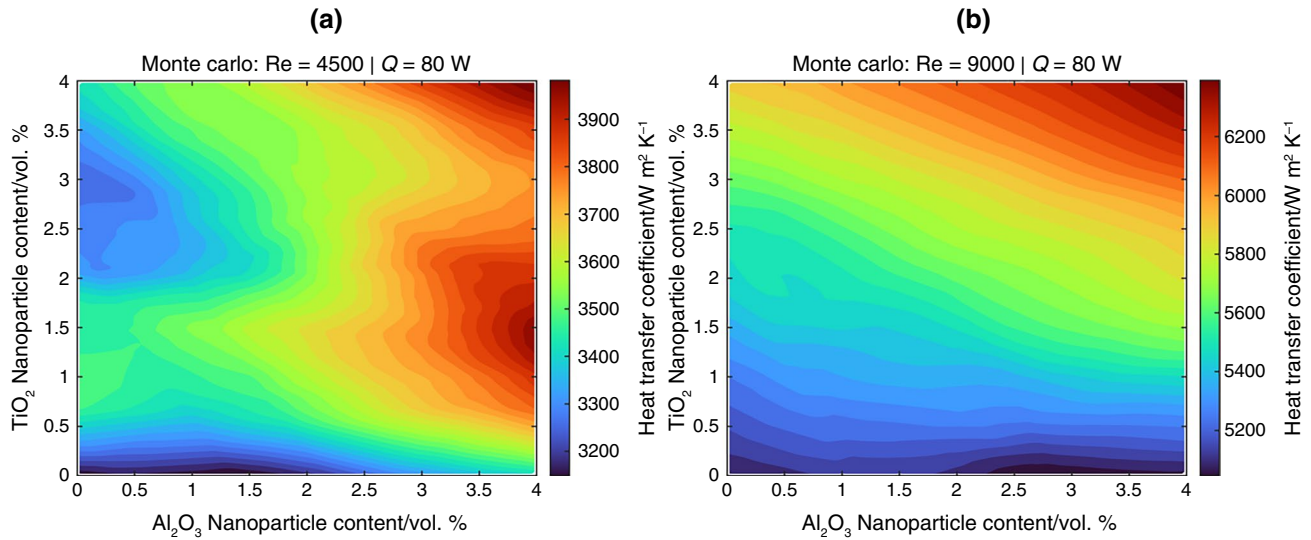


Fig. 7 Monte Carlo-predicted heat transfer coefficient h across the Al_2O_3 - TiO_2 composition plane at $Q = 80$ W: (a) $Re = 4,500$ and (b) $Re = 9,000$

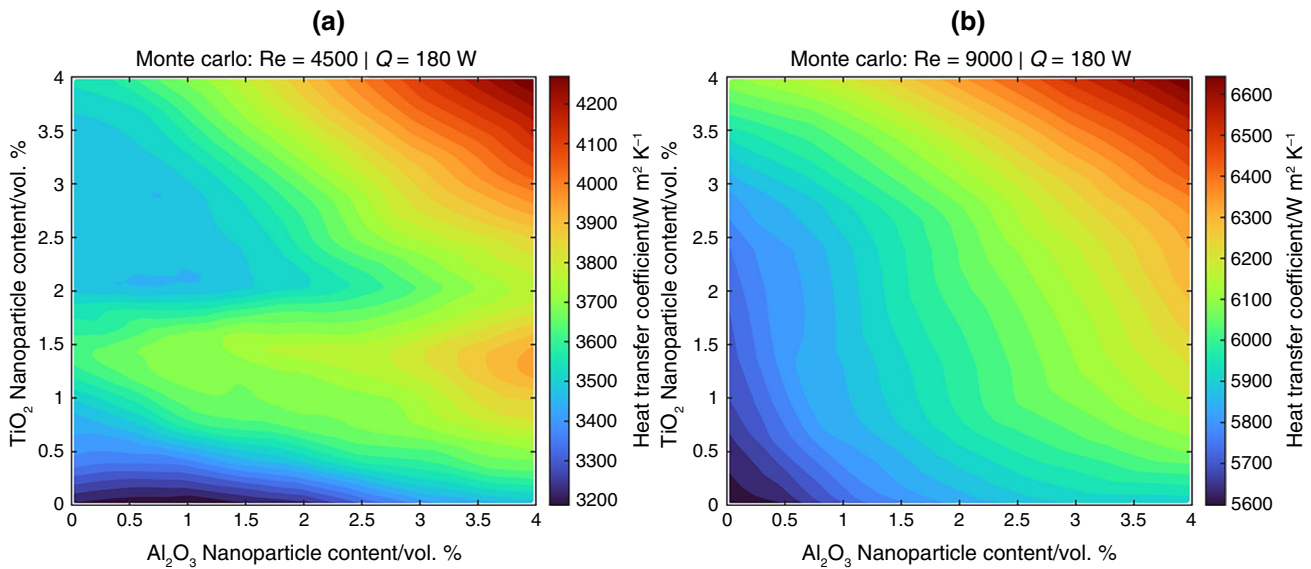


Fig. 8 Monte Carlo-predicted heat transfer coefficient h across the Al_2O_3 - TiO_2 composition plane at $Q = 180$ W: (a) $Re = 4,500$ and (b) $Re = 9,000$

numbers generally produce positive SHAP values, whereas lower Reynolds numbers produce negative values. This behavior is physically consistent with the increase in coolant velocity and mixing intensity, which reduces coolant-side thermal resistance and increases the wall-temperature gradient. Heat-generation rate has a comparatively small influence on the predicted heat-transfer coefficient because, at a fixed flow condition, the imposed heat flux and the resulting wall-to-coolant temperature difference increase together.

Maximum hotspot temperature is controlled mainly by heat-generation rate and Reynolds number. Higher heat-generation rates produce positive SHAP values because the larger thermal load increases the temperature difference required for heat removal. In contrast, higher Reynolds numbers generally produce negative SHAP values by strengthening convection and reducing the battery-to-coolant thermal resistance. The Al_2O_3 and TiO_2 volume fractions have smaller SHAP magnitudes for both outputs, indicating that coolant composition provides a secondary

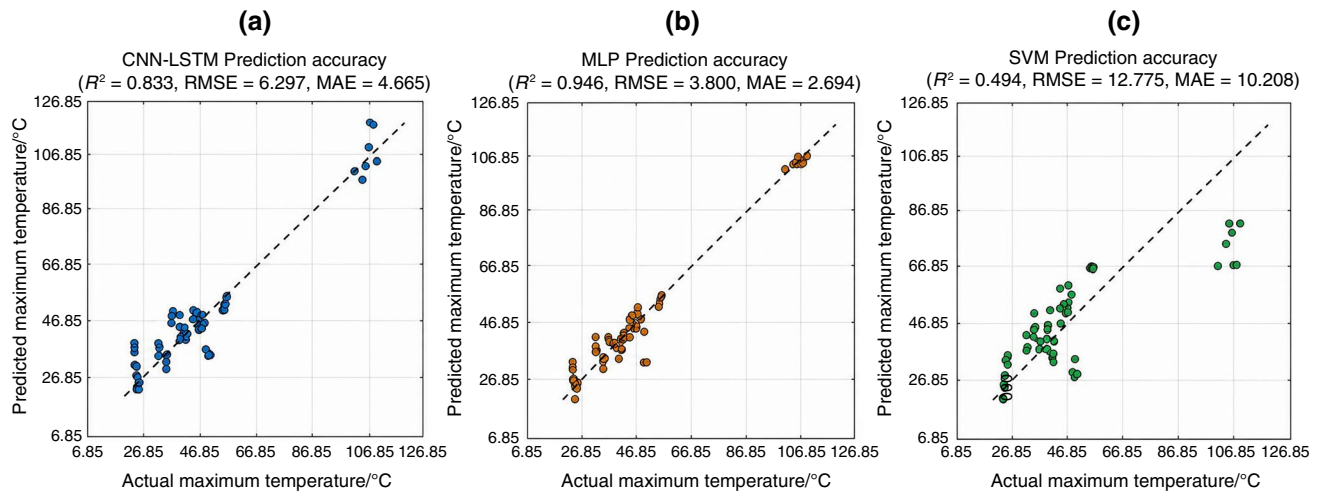


Fig. 9 Cross-validated CFD-derived versus ML-predicted maximum hotspot temperature $T_{\max}$: (a) CNN-LSTM, (b) MLP, and (c) SVR. The dashed line indicates perfect agreement

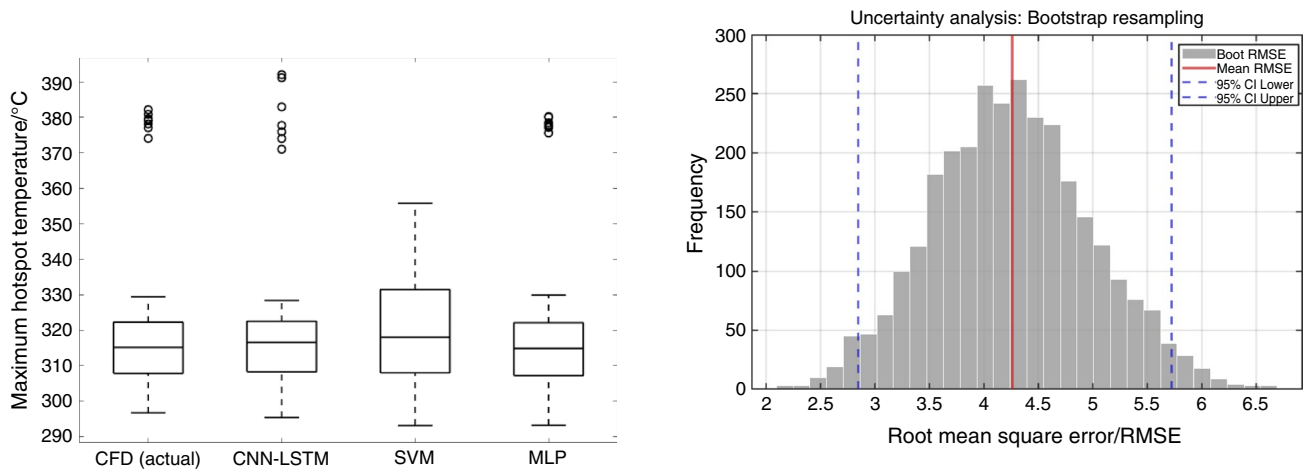


Fig. 10 Boxplot comparison of CFD-derived and ML-predicted maximum hotspot temperature $T_{\max}$ values for CNN-LSTM, SVR, and MLP

correction through changes in effective thermal conductivity and viscosity.

From a BTMS design perspective, these results indicate that coolant flow rate should first be selected to maintain the hotspot temperature below the required limit at the maximum expected thermal load. Nanoparticle concentration may then be adjusted to obtain an incremental thermal improvement, subject to additional consideration of viscosity, pressure drop, pumping power, and nanofluid stability. Because SHAP explains the trained surrogate rather than the underlying CFD model directly, these results represent model-dependent associations and should not be interpreted as causal effects.

Fig. 11 Bootstrap uncertainty distribution for maximum hotspot temperature $T_{\max}$ prediction error. The solid vertical line indicates the mean RMSE, and the dashed vertical lines indicate the 95% confidence interval

Heat-generation effect on coolant temperature ranking

Table 10 summarizes the predicted $T_{\max}$ values at $Re = 4,434$ across the simulated heat-generation range. The nanofluid coolants reduce $T_{\max}$ relative to water at every heat level, with the hybrid nanofluid giving the lowest temperature. However, the composition effect remains small compared with the heat-generation effect. At 5 C, the hybrid nanofluid reduces $T_{\max}$ by 1.6 °C relative to water, while increasing Q from 1 to 5 C raises $T_{\max}$ by more than 30 °C. This confirms that coolant composition improves thermal performance, but it cannot replace sufficient flow-rate control under high heat-generation conditions.

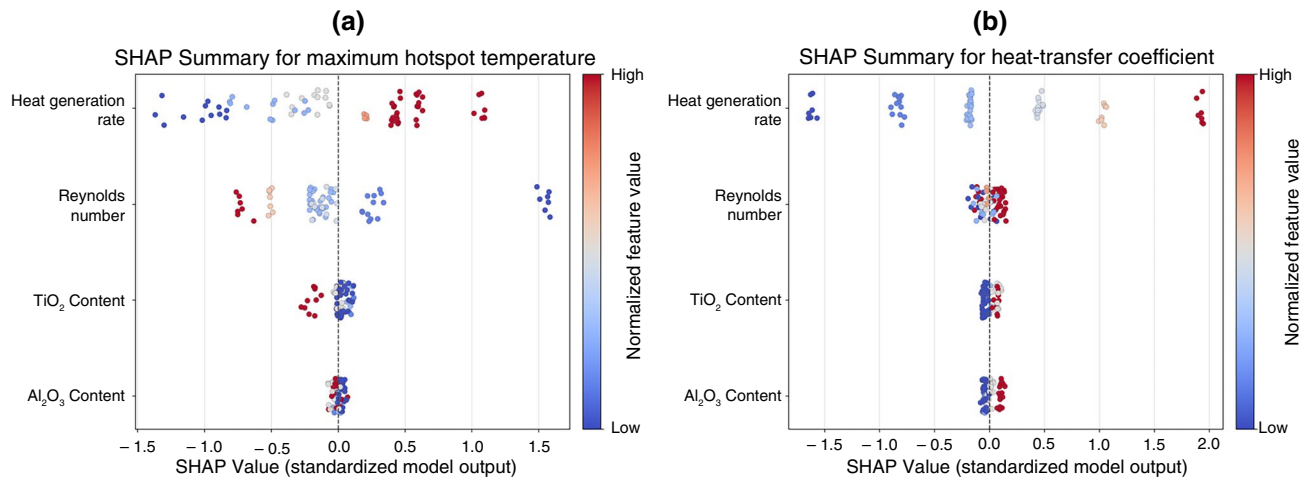


Fig. 12 SHAP beeswarm plots for the MLP predictions of (a) maximum hotspot temperature and (b) heat-transfer coefficient. Positive SHAP values increase the predicted response, whereas negative val-

ues decrease it. Point color indicates the normalized feature value, from low (blue) to high (red)

Table 10 CFD-derived maximum hotspot temperatures at $Re = 4434$ across the simulated heat-generation range

Q/W	Water /°C	Al_2O_3 (4%) /°C	TiO_2 (4%) /°C	Hybrid (4%) /°C
55.0	24.4	24.3	24.3	24.2
111.0	32.4	32.0	32.0	31.9
166.5	40.4	39.7	39.6	39.5
222.0	48.3	47.4	47.3	47.1
277.5	56.3	55.0	54.9	54.7

Monte Carlo mapping of maximum hotspot temperature

Figures 13 and 14 show the Monte Carlo-predicted maximum hotspot temperature across the Al_2O_3 – TiO_2 composition plane at $Q = 80$ W and $Q = 180$ W, respectively. As expected, increasing the heat-generation rate raises the predicted temperature field. At each heat load, increasing Reynolds number improves convective heat removal and reduces the predicted hotspot temperature. The maps also indicate

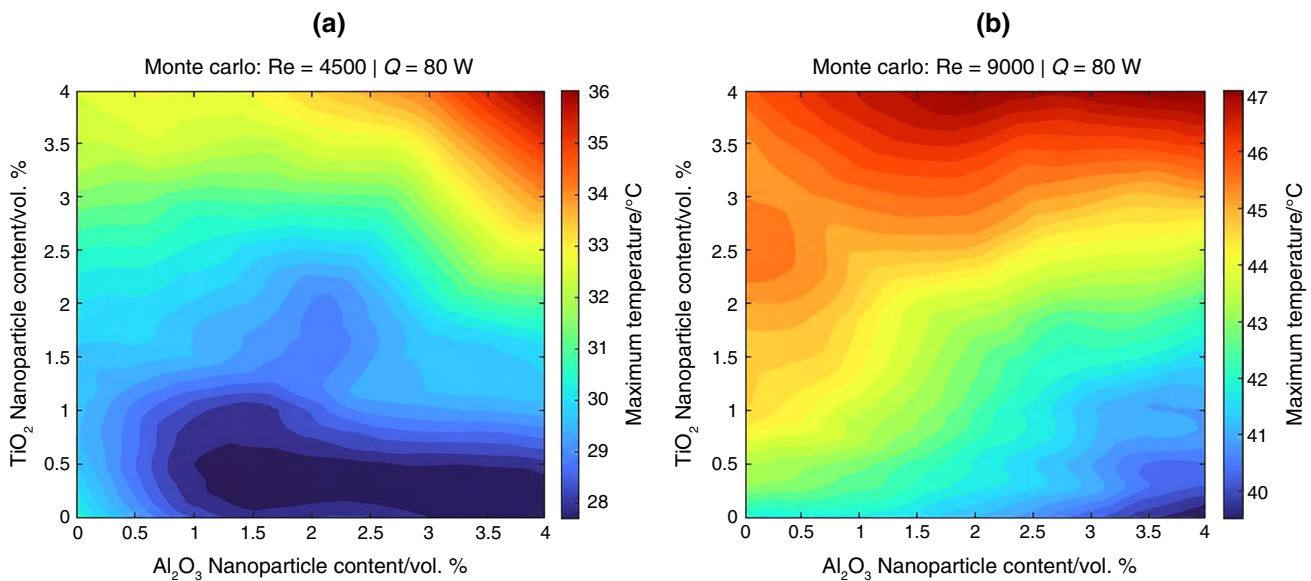


Fig. 13 Monte Carlo-predicted maximum hotspot temperature $T_{\max}$ across the Al_2O_3 – TiO_2 composition plane at $Q = 80$ W: (a) $Re = 4,500$ and (b) $Re = 9,000$

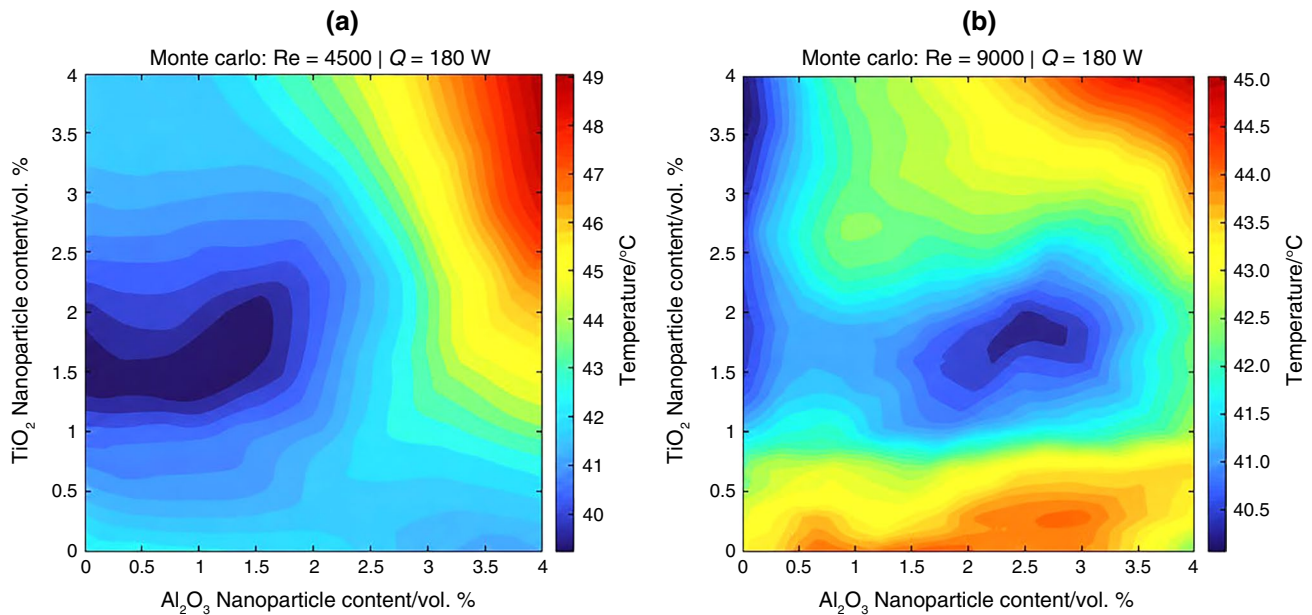


Fig. 14 Monte Carlo-predicted maximum hotspot temperature $T_{\max}$ across the Al_2O_3 – TiO_2 composition plane at $Q = 180 \text{ W}$: (a) $Re = 4,500$ and (b) $Re = 9,000$

that increasing nanoparticle loading can reduce $T_{\max}$, but the magnitude of this improvement is smaller than the changes caused by heat generation rate and Reynolds number.

This supports the main design conclusion that flow rate and thermal load set the primary operating temperature, while nanofluent composition provides a secondary correction. As with the h maps, off-diagonal composition regions should be interpreted carefully because the CFD data mainly validate the fixed 50:50 hybrid path.

The composition maps should be interpreted as a screening tool rather than as evidence of a unique optimum. At high Reynolds number, the lower convective resistance reduces the relative importance of coolant conductivity, and the compositional temperature differences become smaller. At lower Reynolds number, conductivity enhancement has a greater influence because coolant-side resistance represents a larger fraction of the total thermal resistance. In practice, the Reynolds number should first be selected to satisfy the hotspot-temperature limit at the maximum anticipated heat load. Nanofluent composition may then be used for incremental temperature reduction, subject to a separate assessment of viscosity, stability, pressure drop, and pumping power.

Reynolds number primarily controls the heat-transfer coefficient because increasing coolant velocity reduces the coolant-side thermal resistance and strengthens convective heat removal. The maximum hotspot temperature is influenced by both Reynolds number and total thermal load because it depends on the balance between internal battery heat generation and heat removal by the coolant. Nanoparticle loading provides a smaller contribution by increasing the

effective thermal conductivity of the coolant. This contribution becomes less important at higher Reynolds numbers, where forced convection dominates the thermal response. From a design perspective, sufficient coolant flow should first be selected to satisfy the maximum-temperature requirement, after which nanofluent composition may provide an additional temperature reduction.

Conclusions

This study developed machine-learning surrogate models for Al_2O_3 – TiO_2 /water nanofluent cooling of a lithium-ion pouch cell using a 72-case validated CFD dataset. The dataset combined a baseline Reynolds-number and composition matrix with a heat-generation extension from 55 to 277.5 W, allowing the relative effects of flow rate, thermal load, and coolant composition to be evaluated.

Across the combined 72-case dataset, the heat transfer coefficient h is governed almost entirely by Reynolds number: at a fixed Reynolds number, h changed by less than 0.5% over the investigated thermal-load range, consistent with the definition $h = q''/(T_{\text{wall}} - T_{\text{inlet}})$, in which the applied heat flux and the resulting temperature difference scale together with Q . Maximum hotspot temperature was governed jointly by Reynolds number and heat-generation rate. Increasing the Reynolds number enhanced convective heat removal, whereas increasing the thermal load produced a larger battery-to-coolant temperature difference. Coolant composition provides a consistent but secondary correction: the hybrid nanofluent reduced $T_{\max}$ by

0.2 °C relative to water at 55 W and by 1.6 °C at 277.5 W, against a roughly 30 °C swing driven by heat generation rate alone, and the Monte Carlo maps show this compositional benefit narrowing further as Reynolds number increases. For BTMS design, these results indicate that flow rate should be sized against the highest anticipated discharge rate, since its cooling benefit is not fixed but grows with thermal load, while nanofluid composition offers a secondary, Reynolds-dependent refinement rather than a substitute for adequate flow.

Although the expanded 72-case dataset improves the coverage of the original matrix, some limitations remain. The heat-generation extension was performed only at 4% total nanoparticle loading. In addition, the full 55–277.5 W thermal-load range was simulated at $Re = 4,434$, whereas the $Re = 3,200$ and $Re = 5,800$ cases were limited to thermal loads of 55–166.5 W. Therefore, the surrogate predictions should be interpreted within the sampled design space. Future work should extend the matrix to additional nanoparticle loadings and higher heat-generation levels at the outer Reynolds numbers. Even with these limits, the 72-case dataset provides a structured basis for identifying the relative roles of flow rate, thermal load, and coolant composition in nanofluid-based battery thermal management.

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Author contribution A.U. conceived the study, developed the methodology, performed the simulations, validation, formal analysis, investigation, data curation, and visualization, and wrote the original manuscript. I.U. contributed to the methodology, software development, formal analysis, investigation, data curation, visualization, and manuscript revision. E.N.O. contributed to software development, formal analysis, investigation, visualization, and manuscript revision. N.V. contributed to software development, formal analysis, visualization, and manuscript revision. K.C. supervised the study, managed the project, contributed to the methodology and validation, and revised the manuscript. All authors reviewed and approved the final manuscript.

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Data Availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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